

ORF309 S2025 Midterm1

March 2025

Problem 1. Bob is going to take an exam for his Probability course. There are 2 questions. Bob's performances on the two questions are not mutually independent, as, clearly, both depend on his understanding of the course materials. Bob estimates that he has probability 0.8 to answer the first question correctly. Given he answers the first question correctly, he has probability 0.6 to answer the second question correctly. If he gets the first question wrong, then he has probability only 0.1 to answer the second question correctly.

(a) What is the probability that Bob answers the second question correctly?

(b) Suppose we know that Bob answers the second question correctly. Given this information, what is the probability that he answers the first question correctly?

(c) Each question is worth 10 points. Bob will get the points if he answers a question correctly, and no point otherwise. What is the variance of Bob's grade?

Solution. (a) Let A denote the event that Bob answers the first question correctly, and B denote the event that Bob answers the second question correctly. We use the Law of Total Probability to evaluate

$$\mathbb{P}[B] = \mathbb{P}[B|A]\mathbb{P}[A] + \mathbb{P}[B|A^c]\mathbb{P}[A^c] = 0.6 * 0.8 + 0.1 * 0.2 = 0.5.$$

(b) By the definition of conditional probability,

$$\mathbb{P}[A|B] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} = \frac{\mathbb{P}[B|A]\mathbb{P}[A]}{\mathbb{P}[B]} = \frac{0.6 * 0.8}{0.5} = 0.96.$$

(c) Let X be Bob's grade. Then

$$X = 10 \cdot (\mathbf{1}_A + \mathbf{1}_B).$$

By the linearity of expectation,

$$\mathbb{E}[X] = 10 \cdot (\mathbb{E}[\mathbf{1}_A] + \mathbb{E}[\mathbf{1}_B]) = 10 \cdot (\mathbb{P}[A] + \mathbb{P}[B]) = 10 * (0.8 + 0.5) = 13,$$

and

$$\begin{aligned}\mathbb{E}[X^2] &= 10^2 \cdot \mathbb{E}[\mathbf{1}_A + \mathbf{1}_B + 2\mathbf{1}_A\mathbf{1}_B] = 10^2 \cdot (\mathbb{P}[A] + \mathbb{P}[B] + 2\mathbb{P}[A \cap B]) \\ &= 10^2 * (0.8 + 0.5 + 2 * 0.6 * 0.8) = 226.\end{aligned}$$

Finally,

$$\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 226 - 13^2 = 57.$$

□

Problem 2. Trains at Princeton Junction Station often experience unpredictable delays. Suppose that the delay times (in minutes) of the first scheduled trains in the afternoon to New York and Trenton are both geometrically distributed, with parameters $p = 0.1$ and $q = 0.2$, respectively (as the clock at the train station only shows minutes, not seconds, we model the delay times as discrete random variables). The delays of the two trains are probabilistically independent.

(a) Suppose that we know the train to New York is delayed by a time strictly longer than 10 minutes. Given this information, what is the probability that the total delay time of that train is strictly longer than 20 minutes?

(b) Suppose we know that the train to Trenton is delayed by 5 minutes. Given this information, how long do we expect the train to New York to be delayed?

(c) Calculate the probability that the train to New York is delayed by a strictly shorter time than the train to Trenton.

Solution. Let X, Y be the delays for the trains to New York and Trenton, respectively.

(a) Since X is geometrically distributed with parameter $p = 0.1$, we know that for any $m \in \{0, 1, 2, \dots\}$,

$$\mathbb{P}[X > m] = (1 - p)^m.$$

The desired conditional probability is thus evaluated as

$$\mathbb{P}[X > 20 | X > 10] = \frac{\mathbb{P}[X > 20]}{\mathbb{P}[X > 10]} = \frac{(1 - p)^{20}}{(1 - p)^{10}} = (1 - p)^{10} = 0.9^{10}.$$

(b) As X, Y are independent, the conditional expectation reduces to the unconditional expectation:

$$\mathbb{E}[X | Y = 5] = \mathbb{E}[X] = \frac{1}{p} = 10.$$

(c) Since Y is geometrically distributed with parameter $q = 0.2$, we know that for any $m \in \{0, 1, 2, \dots\}$,

$$\mathbb{P}[Y > m] = (1 - q)^m.$$

Now we can condition on X , apply the law of total probabilities, and use the independence between X and Y to get that

$$\begin{aligned} \mathbb{P}[X < Y] &= \sum_{m=1}^{\infty} \mathbb{P}[X < Y | X = m] \mathbb{P}[X = m] \\ &= \sum_{m=1}^{\infty} \mathbb{P}[Y > m] \mathbb{P}[X = m] = \sum_{m=1}^{\infty} (1 - q)^m (1 - p)^{m-1} p = \frac{p(1 - q)}{p + q - pq} = \frac{2}{7}. \end{aligned}$$

□

Problem 3. Alice invests in Fruity Yogurt Inc. by holding several shares of its stock and cares about the evolution of the stock price. The stock price of Fruity Yogurt Inc. follows a mysteriously discrete pattern. On each day, the stock price only changes once, **exactly at noon**; the stock price will be either doubled (with probability 0.4) or halved (with probability 0.6). Suppose that the price movements of different days are mutually independent. **After** the noon of today (which represents day 0), Fruity Yogurt's stock price is \$1.

(a) Calculate the probability that **after** the noon of day 5, the stock's price is \$2. (Hint: think about how many days out of the 5 days should the price go up?)

(b) If n is very large number, estimate the fraction of trading days (among the first n trading days) on which the stock price is doubled.

(c) Calculate the expectation of the stock price **after** the noon of day n . What happens as $n \rightarrow \infty$?

Solution. Let S_n be the stock price after n trading days. Define

$$X_n := \begin{cases} 2 & \text{if the stock price is doubled on day } n \\ \frac{1}{2} & \text{if the stock price is halved on day } n. \end{cases}$$

Then $X_1, \dots, X_n$ are i.i.d. with $\mathbb{P}[X_i = 2] = 0.4$, $\mathbb{P}[X_i = \frac{1}{2}] = 0.6$. Let

$$U_n := \sum_{i=1}^n \mathbf{1}_{\{X_i=2\}}$$

be the number of days in which the stock price is doubled, out of the first n trading days. We see that U_n is Binomial(n, p) distributed with $p = 0.4$.

(a) In order for the stock to be traded at 2 dollar per share after 5 days, there must be exactly 3 days in which the price is doubled. With the above analysis,

$$\mathbb{P}[S_5 = 2] = \mathbb{P}[U_5 = 3] = \binom{5}{3} 0.4^3 0.6^2.$$

(b) By the Law of Large Numbers, $\frac{U_n}{n}$ converges to 0.4 as $n \rightarrow \infty$.

(c) First, it is easy to calculate

$$\mathbb{E}[X_1] = 2 * \mathbb{P}[X_1 = 2] + \frac{1}{2} * \mathbb{P}[X_1 = \frac{1}{2}] = 2 * 0.4 + \frac{1}{2} * 0.6 = 1.1.$$

Note that we can express S_n as

$$S_n = \prod_{i=1}^n X_i.$$

By the mutual independence of $X_1, \dots, X_n$,

$$\mathbb{E}[S_n] = \mathbb{E} \left[\prod_{i=1}^n X_i \right] = \prod_{i=1}^n \mathbb{E}[X_i] = 1.1^n.$$

The above tends to ∞ as $n \rightarrow \infty$.

□